AEMP Dashboard Folder Structure

Administrative & Enterprise Management Platform

aemp-dashboard.btpma.org

Complete Technical Documentation
Directory Architecture, API Routes, Components & Configuration

BTPMA — Black Trust Property Management Association
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1. Project Overview

The **AEMP Dashboard** is the administrative backbone of the BTPMA ecosystem, hosted at `aemp-dashboard.btpma.org`. Built with **Next.js 14+ (App Router)**, it provides a comprehensive management interface for administrators, staff, and authorized personnel to manage members, trusts, education content, AI configurations, and platform-wide settings.

> **Tech Stack:** Next.js 14 (App Router) • React 18 • TypeScript • Tailwind CSS • Prisma ORM • PostgreSQL • NextAuth.js • Shadcn/UI

Key Responsibilities

- **Member Management:** CRUD operations for all BTPMA members, roles, and permissions
- **Trust Administration:** Estate trust directive management and document generation
- **Education Management:** Course creation, module editing, and content publishing
- **AI Configuration:** Head Steward AI training data, prompt management, and analytics
- **Financial Oversight:** Payment tracking, subscription management, and reporting
- **Content Management:** Blog posts, announcements, and public website content
- **Analytics Dashboard:** Platform-wide metrics, user engagement, and system health

2. Complete Folder Structure

```
aemp-dashboard.btpma.org/
├── .env                        # Environment variables
├── .env.local                  # Local overrides (git-ignored)
├── .eslintrc.json              # ESLint configuration
├── .gitignore                  # Git ignore rules
├── .prettierrc                 # Prettier formatting config
├── next.config.js              # Next.js configuration
├── package.json                # Dependencies & scripts
├── postcss.config.js           # PostCSS / Tailwind config
├── tailwind.config.ts          # Tailwind CSS configuration
├── tsconfig.json               # TypeScript configuration
├── prisma/
│   ├── schema.prisma           # Database schema
│   ├── seed.ts                 # Database seeder
│   └── migrations/             # Database migrations
│       ├── 001_init/
│       ├── 002_add_trusts/
│       └── 003_add_education/
├── public/
│   ├── favicon.ico
```

```
│   ├── logo.svg
│   ├── logo-dark.svg
│   ├── images/
│   │   ├── dashboard/
│   │   ├── avatars/
│   │   └── placeholders/
│   └── fonts/
│       ├── inter-var.woff2
│       └── geist-mono.woff2
├── src/
│   ├── app/
│   │   ├── layout.tsx            # Root layout (providers, auth)
│   │   ├── page.tsx              # Dashboard home / redirect
│   │   ├── loading.tsx           # Global loading state
│   │   ├── error.tsx             # Global error boundary
│   │   ├── not-found.tsx         # 404 page
│   │   ├── globals.css           # Global styles
│   │   ├── (auth)/
│   │   │   ├── login/
│   │   │   │   └── page.tsx      # Admin login
│   │   │   ├── forgot-password/
│   │   │   │   └── page.tsx
│   │   │   └── layout.tsx        # Auth layout (no sidebar)
│   │   ├── (dashboard)/
│   │   │   ├── layout.tsx        # Dashboard layout (sidebar+header)
│   │   │   ├── overview/
│   │   │   │   └── page.tsx      # Main dashboard overview
│   │   │   ├── members/
│   │   │   │   ├── page.tsx      # Members list
│   │   │   │   ├── [id]/
│   │   │   │   │   ├── page.tsx  # Member detail
│   │   │   │   │   └── edit/
│   │   │   │   │       └── page.tsx
│   │   │   │   └── new/
│   │   │   │       └── page.tsx  # Add new member
│   │   │   ├── trusts/
│   │   │   │   ├── page.tsx      # Trust directives list
│   │   │   │   ├── [id]/
│   │   │   │   │   └── page.tsx  # Trust detail
│   │   │   │   └── templates/
│   │   │   │       └── page.tsx  # Trust templates
│   │   │   ├── education/
│   │   │   │   ├── page.tsx      # Education overview
│   │   │   │   ├── courses/
│   │   │   │   │   ├── page.tsx  # Course list
│   │   │   │   │   ├── [id]/
│   │   │   │   │   │   ├── page.tsx
│   │   │   │   │   │   └── edit/
│   │   │   │   │   │       └── page.tsx
│   │   │   │   │   └── new/
│   │   │   │   │       └── page.tsx
│   │   │   │   └── modules/
│   │   │   │       ├── page.tsx
│   │   │   │       └── [id]/
```

```
│   │   │   │           └── page.tsx
│   │   │   ├── ai-config/
│   │   │   │   ├── page.tsx      # AI configuration
│   │   │   │   ├── prompts/
│   │   │   │   │   └── page.tsx  # Prompt management
│   │   │   │   ├── training/
│   │   │   │   │   └── page.tsx  # Training data
│   │   │   │   └── analytics/
│   │   │   │       └── page.tsx  # AI usage analytics
│   │   │   ├── finances/
│   │   │   │   ├── page.tsx      # Financial overview
│   │   │   │   ├── payments/
│   │   │   │   │   └── page.tsx
│   │   │   │   ├── subscriptions/
│   │   │   │   │   └── page.tsx
│   │   │   │   └── reports/
│   │   │   │       └── page.tsx
│   │   │   ├── content/
│   │   │   │   ├── page.tsx      # Content management
│   │   │   │   ├── blog/
│   │   │   │   │   ├── page.tsx
│   │   │   │   │   ├── [slug]/
│   │   │   │   │   │   └── page.tsx
│   │   │   │   │   └── new/
│   │   │   │   │       └── page.tsx
│   │   │   │   └── announcements/
│   │   │   │       └── page.tsx
│   │   │   ├── settings/
│   │   │   │   ├── page.tsx      # General settings
│   │   │   │   ├── profile/
│   │   │   │   │   └── page.tsx
│   │   │   │   ├── security/
│   │   │   │   │   └── page.tsx
│   │   │   │   └── integrations/
│   │   │   │       └── page.tsx
│   │   │   └── analytics/
│   │   │       ├── page.tsx      # Analytics dashboard
│   │   │       ├── users/
│   │   │       │   └── page.tsx
│   │   │       └── engagement/
│   │   │           └── page.tsx
│   │   └── api/
│   │       ├── auth/
│   │       │   ├── [...nextauth]/
│   │       │   │   └── route.ts  # NextAuth handler
│   │       │   └── register/
│   │       │       └── route.ts
│   │       ├── members/
│   │       │   ├── route.ts      # GET all, POST new
│   │       │   └── [id]/
│   │       │       └── route.ts  # GET, PUT, DELETE
│   │       ├── trusts/
│   │       │   ├── route.ts
│   │       │   └── [id]/
```

```
│   │       │       ├── route.ts
│   │       │       └── generate/
│   │       │           └── route.ts  # PDF generation
│   │       ├── education/
│   │       │   ├── courses/
│   │       │   │   ├── route.ts
│   │       │   │   └── [id]/
│   │       │   │       └── route.ts
│   │       │   └── modules/
│   │       │       ├── route.ts
│   │       │       └── [id]/
│   │       │           └── route.ts
│   │       ├── ai/
│   │       │   ├── chat/
│   │       │   │   └── route.ts  # AI chat endpoint
│   │       │   ├── prompts/
│   │       │   │   └── route.ts
│   │       │   └── analytics/
│   │       │       └── route.ts
│   │       ├── finances/
│   │       │   ├── payments/
│   │       │   │   └── route.ts
│   │       │   └── subscriptions/
│   │       │       └── route.ts
│   │       ├── content/
│   │       │   ├── blog/
│   │       │   │   ├── route.ts
│   │       │   │   └── [slug]/
│   │       │   │       └── route.ts
│   │       │   └── announcements/
│   │       │       └── route.ts
│   │       ├── upload/
│   │       │   └── route.ts      # File upload handler
│   │       ├── analytics/
│   │       │   └── route.ts
│   │       └── webhooks/
│   │           ├── stripe/
│   │           │   └── route.ts  # Stripe webhooks
│   │           └── sendgrid/
│   │               └── route.ts
│   ├── components/
│   │   ├── ui/                   # Shadcn/UI primitives
│   │   │   ├── button.tsx
│   │   │   ├── card.tsx
│   │   │   ├── dialog.tsx
│   │   │   ├── dropdown-menu.tsx
│   │   │   ├── input.tsx
│   │   │   ├── label.tsx
│   │   │   ├── select.tsx
│   │   │   ├── table.tsx
│   │   │   ├── tabs.tsx
│   │   │   ├── textarea.tsx
│   │   │   ├── toast.tsx
│   │   │   ├── tooltip.tsx
```

```
│   │   │   └── badge.tsx
│   │   ├── layout/
│   │   │   ├── sidebar.tsx       # Dashboard sidebar
│   │   │   ├── header.tsx        # Dashboard header
│   │   │   ├── footer.tsx
│   │   │   ├── breadcrumbs.tsx
│   │   │   └── mobile-nav.tsx
│   │   ├── dashboard/
│   │   │   ├── stats-cards.tsx
│   │   │   ├── recent-activity.tsx
│   │   │   ├── charts/
│   │   │   │   ├── line-chart.tsx
│   │   │   │   ├── bar-chart.tsx
│   │   │   │   ├── pie-chart.tsx
│   │   │   │   └── area-chart.tsx
│   │   │   └── quick-actions.tsx
│   │   ├── members/
│   │   │   ├── member-table.tsx
│   │   │   ├── member-form.tsx
│   │   │   ├── member-card.tsx
│   │   │   └── member-filters.tsx
│   │   ├── trusts/
│   │   │   ├── trust-form.tsx
│   │   │   ├── trust-viewer.tsx
│   │   │   └── trust-template-selector.tsx
│   │   ├── education/
│   │   │   ├── course-editor.tsx
│   │   │   ├── module-editor.tsx
│   │   │   ├── content-block.tsx
│   │   │   └── media-uploader.tsx
│   │   ├── ai/
│   │   │   ├── prompt-editor.tsx
│   │   │   ├── chat-preview.tsx
│   │   │   └── training-data-table.tsx
│   │   ├── finances/
│   │   │   ├── payment-table.tsx
│   │   │   ├── revenue-chart.tsx
│   │   │   └── subscription-manager.tsx
│   │   └── shared/
│   │       ├── data-table.tsx    # Reusable data table
│   │       ├── file-upload.tsx
│   │       ├── rich-text-editor.tsx
│   │       ├── search-input.tsx
│   │       ├── pagination.tsx
│   │       ├── confirm-dialog.tsx
│   │       ├── status-badge.tsx
│   │       └── empty-state.tsx
│   ├── lib/
│   │   ├── prisma.ts             # Prisma client singleton
│   │   ├── auth.ts               # NextAuth configuration
│   │   ├── utils.ts              # General utilities
│   │   ├── validations.ts        # Zod schemas
│   │   ├── constants.ts          # App constants
│   │   ├── api-client.ts         # API helper functions
```

```
│   │   ├── stripe.ts            # Stripe configuration
│   │   ├── email.ts             # Email service (SendGrid)
│   │   ├── ai.ts                # OpenAI / AI SDK config
│   │   ├── upload.ts            # File upload utilities
│   │   └── pdf-generator.ts     # Trust PDF generation
│   ├── hooks/
│   │   ├── use-auth.ts
│   │   ├── use-members.ts
│   │   ├── use-debounce.ts
│   │   ├── use-pagination.ts
│   │   └── use-toast.ts
│   ├── types/
│   │   ├── index.ts             # Shared type exports
│   │   ├── member.ts
│   │   ├── trust.ts
│   │   ├── education.ts
│   │   ├── finance.ts
│   │   └── api.ts
│   ├── stores/
│   │   ├── auth-store.ts        # Zustand auth store
│   │   ├── sidebar-store.ts
│   │   └── notification-store.ts
│   └── middleware.ts            # Auth & domain middleware
└── docker/
    ├── Dockerfile
    ├── docker-compose.yml
    └── nginx.conf
```

3. Root Configuration Files

File	Purpose
`next.config.js`	Next.js settings: image domains, redirects, env vars, webpack overrides
`tailwind.config.ts`	Custom theme colors, fonts, breakpoints, and plugin configuration
`tsconfig.json`	TypeScript compiler options with path aliases (`@/` → `src/`)
`prisma/schema.prisma`	Database models: User, Member, Trust, Course, Module, Payment, etc.
`.env`	DATABASE_URL, NEXTAUTH_SECRET, STRIPE_KEY, OPENAI_KEY, SENDGRID_KEY
`package.json`	Scripts: `dev` , `build` , `start` , `db:push` , `db:seed` , `lint`

4. Source Directory — `src/`

The `src/` directory contains all application source code, organized by concern:

Directory	Purpose
`src/app/`	Next.js App Router — pages, layouts, API routes
`src/components/`	Reusable React components organized by feature domain
`src/lib/`	Shared utilities, service configurations, and helper functions
`src/hooks/`	Custom React hooks for data fetching and UI logic
`src/types/`	TypeScript type definitions and interfaces
`src/stores/`	Zustand state management stores
`src/middleware.ts`	Request middleware for authentication and domain routing

5. Application Router — `src/app/`

Route Groups

Next.js route groups (parenthesized folders) organize routes without affecting the URL path:

- `(auth)/` — Authentication pages with a minimal layout (no sidebar). Includes login and forgot-password flows.
- `(dashboard)/` — All authenticated dashboard pages sharing the sidebar + header layout.
- `api/` — Server-side API route handlers for all CRUD operations and integrations.

Dashboard Modules

Route	Module	Description
`/overview`	Overview	Main dashboard with KPIs, charts, and recent activity
`/members`	Members	Member directory, profiles, role management
`/trusts`	Trusts	Estate trust directives, templates, PDF generation
`/education`	Education	Course & module management, content editing
`/ai-config`	AI Config	Head Steward AI prompts, training data, analytics

`/finances`	Finances	Payments, subscriptions, revenue reports
`/content`	Content	Blog posts, announcements, CMS
`/settings`	Settings	Profile, security, integrations
`/analytics`	Analytics	User engagement, platform metrics

6. API Routes Documentation

Authentication

Method	Endpoint	Description
POST	`/api/auth/[...nextauth]`	NextAuth.js handler (login, session, callbacks)
POST	`/api/auth/register`	Admin user registration (invite-only)

Members API

Method	Endpoint	Description
GET	`/api/members`	List all members (paginated, filterable)
POST	`/api/members`	Create a new member
GET	`/api/members/[id]`	Get member by ID
PUT	`/api/members/[id]`	Update member details
DELETE	`/api/members/[id]`	Delete / deactivate member

Trusts API

Method	Endpoint	Description
GET	`/api/trusts`	List all trust directives
POST	`/api/trusts`	Create new trust directive
GET	`/api/trusts/[id]`	Get trust by ID
POST	`/api/trusts/[id]/generate`	Generate trust PDF document

Education API

Method	Endpoint	Description
GET/POST	`/api/education/courses`	List / create courses
GET/PUT/DELETE	`/api/education/courses/[id]`	Course CRUD
GET/POST	`/api/education/modules`	List / create modules
GET/PUT/DELETE	`/api/education/modules/[id]`	Module CRUD

AI, Finances, Content & Webhooks

Method	Endpoint	Description
POST	`/api/ai/chat`	AI chat completion (streaming)
GET/POST	`/api/ai/prompts`	Manage AI system prompts
GET	`/api/ai/analytics`	AI usage statistics
GET	`/api/finances/payments`	Payment history
GET	`/api/finances/subscriptions`	Active subscriptions
POST	`/api/upload`	File upload (images, documents)
POST	`/api/webhooks/stripe`	Stripe payment webhooks
POST	`/api/webhooks/sendgrid`	Email delivery webhooks

7. Component Organization

UI Primitives — `components/ui/`

Base components from **Shadcn/UI**, customized with BTPMA brand colors. These are atomic, unstyled-by-default components that accept variants and className overrides.

Layout Components — `components/layout/`

- **sidebar.tsx** — Collapsible navigation with role-based menu items, active state indicators
- **header.tsx** — Top bar with search, notifications bell, user avatar dropdown
- **breadcrumbs.tsx** — Auto-generated breadcrumb trail from route segments
- **mobile-nav.tsx** — Responsive hamburger menu for tablet/mobile views

Feature Components

Each dashboard module has its own component directory containing forms, tables, cards, and specialized UI for that domain. Components are co-located with their feature for maintainability.

Shared Components — `components/shared/`

Cross-cutting components used across multiple modules: data tables with sorting/filtering, file uploaders, rich text editors, pagination controls, and confirmation dialogs.

8. Shared Utilities & Libraries — `src/lib/`

File	Purpose
`prisma.ts`	Singleton Prisma client instance (prevents hot-reload connection leaks)
`auth.ts`	NextAuth config: providers (credentials, Google), callbacks, session strategy
`utils.ts`	cn() class merger, formatDate(), formatCurrency(), slugify()
`validations.ts`	Zod schemas for form validation: memberSchema, trustSchema, courseSchema
`api-client.ts`	Typed fetch wrapper with error handling, auth headers, and retry logic
`stripe.ts`	Stripe SDK initialization, helper functions for checkout and subscriptions
`email.ts`	SendGrid email templates: welcome, password reset, trust notification
`ai.ts`	OpenAI SDK config, streaming helpers, token counting utilities
`pdf-generator.ts`	Puppeteer/React-PDF trust document generation pipeline

9. Public Assets — `public/`

- **favicon.ico / logo.svg** — BTPMA brand assets in light and dark variants
- **images/dashboard/** — Dashboard-specific illustrations and icons
- **images/avatars/** — Default avatar placeholders
- **fonts/** — Self-hosted Inter and Geist Mono variable fonts for performance

10. Domain-Aware Routing

The middleware at `src/middleware.ts` inspects the incoming request hostname to enforce domain-specific access control:

```
// src/middleware.ts
import { NextResponse } from 'next/server';
import type { NextRequest } from 'next/server';
import { getToken } from 'next-auth/jwt';

export async function middleware(request: NextRequest) {
  const hostname = request.headers.get('host') || '';
  const pathname = request.nextUrl.pathname;

  // Only allow access from aemp-dashboard.btpma.org
  if (!hostname.includes('aemp-dashboard.btpma.org')
      && !hostname.includes('localhost')) {
    return NextResponse.redirect('https://btpma.org');
  }

  // Protect all dashboard routes
  if (pathname.startsWith('/(dashboard)') || pathname.startsWith('/overview')) {
    const token = await getToken({ req: request });
    if (!token) {
      return NextResponse.redirect(
        new URL('/login', request.url)
      );
    }
    // Role-based access control
    if (pathname.startsWith('/ai-config') && token.role !== 'ADMIN') {
      return NextResponse.redirect(
        new URL('/overview', request.url)
      );
    }
  }

  return NextResponse.next();
}

export const config = {
  matcher: ['/((?!api|_next/static|_next/image|favicon.ico).*)'],
};
```

> **Key Behavior:** Requests not originating from `aemp-dashboard.btpma.org` are redirected to the public site. All dashboard routes require authentication. Sensitive routes like AI configuration require the ADMIN role.

11. Deployment Structure

```
/var/www/
├── aemp-dashboard.btpma.org/     # Admin dashboard (Next.js)
│   ├── .next/                    # Production build output
│   ├── node_modules/
│   ├── public/
│   ├── prisma/
│   └── package.json
├── platform.btpma.org/           # Member portal (Next.js)
│   ├── .next/
│   └── ...
├── btpma.org/                    # Public website
│   └── ...
└── shared/
    ├── ssl/                      # SSL certificates
    │   ├── btpma.org.pem
    │   └── btpma.org.key
    └── logs/
        ├── aemp-access.log
        ├── platform-access.log
        └── public-access.log
```

12. Nginx Configuration

```
# /etc/nginx/sites-available/aemp-dashboard.btpma.org

upstream aemp_dashboard {
    server 127.0.0.1:3001;  # Next.js on port 3001
}

server {
    listen 80;
    server_name aemp-dashboard.btpma.org;
    return 301 https://$server_name$request_uri;
}

server {
    listen 443 ssl http2;
    server_name aemp-dashboard.btpma.org;

    ssl_certificate     /var/www/shared/ssl/btpma.org.pem;
    ssl_certificate_key /var/www/shared/ssl/btpma.org.key;
    ssl_protocols       TLSv1.2 TLSv1.3;
    ssl_ciphers         HIGH:!aNULL:!MD5;

    # Security headers
```

```
    add_header X-Frame-Options "SAMEORIGIN" always;
    add_header X-Content-Type-Options "nosniff" always;
    add_header X-XSS-Protection "1; mode=block" always;
    add_header Strict-Transport-Security
               "max-age=31536000; includeSubDomains" always;

    # Restrict access to admin IPs (optional)
    # allow 203.0.113.0/24;
    # deny all;

    # Static assets with long cache
    location /_next/static/ {
        proxy_pass http://aemp_dashboard;
        expires 365d;
        add_header Cache-Control "public, immutable";
    }

    # API routes — no caching
    location /api/ {
        proxy_pass http://aemp_dashboard;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection "upgrade";
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_read_timeout 120s;
    }

    # All other routes
    location / {
        proxy_pass http://aemp_dashboard;
        proxy_http_version 1.1;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }

    access_log /var/www/shared/logs/aemp-access.log;
    error_log  /var/www/shared/logs/aemp-error.log;
}
```

> **Port Allocation:** aemp-dashboard runs on port **3001**, platform on **3002**, and the public site on **3000**. Each subdomain's Nginx server block proxies to the corresponding port.